Practical 14: Admin Dashboard with Role-Based Access Control, Dynamic Menu, and Audit Log

Problem Definition
Develop an admin dashboard that displays user statistics, student count, event count and latest registrations. Restrict access to admin users only and show dynamic menu options based on role.

Key questions

1. Are dashboard values loaded dynamically from the database?
Yes, dashboard values are loaded dynamically from the database.

2. Is access restricted to admin users?
Yes, access is restricted to admin users.

3. Are role-based menus working?
Yes, role-based menus work according to user roles.

4. Are important admin actions recorded in audit_logs?
Yes, important admin actions are recorded in the `audit_logs` table.

Code
Db.php

```php
<?php
$conn = mysqli_connect("localhost", "root", "", "admin_db");

if (!$conn) {
    die("Database connection failed");
}

mysqli_set_charset($conn, "utf8mb4");

if (session_status() === PHP_SESSION_NONE) {
    session_start();
}
?>
```

Dashboard.php

```php
<?php
include 'db.php';
```

```php
include 'access.php';

// Dashboard Statistics
function getCount($conn, $table) {
    $allowedTables = ['users', 'students', 'events'];

    if (!in_array($table, $allowedTables)) {
        return 0;
    }

    $result = mysqli_query(
        $conn,
        "SELECT COUNT(*) AS total FROM $table"
    );

    if (!$result) {
        return 0;
    }

    $row = mysqli_fetch_assoc($result);
    return $row['total'];
}

$totalUsers = getCount($conn, 'users');
$totalStudents = getCount($conn, 'students');
$totalEvents = getCount($conn, 'events');

// Latest Registrations
$registrations = mysqli_query(
    $conn,
    "SELECT id, name, email, role, created_at
     FROM users
     ORDER BY id DESC
     LIMIT 5"
);

// Recent Audit Logs
$auditLogs = mysqli_query(
    $conn,
    "SELECT action, created_at
```

```
    FROM audit_logs
    ORDER BY id DESC
    LIMIT 5"
);
?>

<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Admin Dashboard</title>

    <link rel="stylesheet" href="style.css">

    <style>
        body {
            font-family: Arial, sans-serif;
            background: #f1f5f9;
            margin: 0;
            padding: 20px;
        }

        .dashboard {
            max-width: 1100px;
            margin: auto;
            background: white;
            padding: 25px;
            border-radius: 10px;
        }

        nav {
            background: #1e293b;
            padding: 15px;
            margin: 20px 0;
            border-radius: 5px;
        }
```

```
        nav a {
            color: white;
            text-decoration: none;
            margin-right: 20px;
        }

        .cards {
            display: flex;
            gap: 20px;
            margin: 25px 0;
            flex-wrap: wrap;
        }

        .card {
            flex: 1;
            min-width: 180px;
            background: #dbeafe;
            padding: 20px;
            border-radius: 8px;
            text-align: center;
        }

        table {
            width: 100%;
            border-collapse: collapse;
            margin: 15px 0 30px;
            background: white;
        }

        th, td {
            border: 1px solid #ddd;
            padding: 12px;
            text-align: left;
        }

        th {
            background: #2563eb;
            color: white;
        }
```

```
        h2 {
            margin-top: 30px;
        }
    </style>
</head>

<body>

<div class="dashboard">

    <h1>Admin Dashboard</h1>

    <p>
        Welcome,
        <?php echo htmlspecialchars($_SESSION['name']); ?>!
    </p>

    <!-- Dynamic Menu -->
    <nav>

        <?php if ($_SESSION['role'] === 'admin') { ?>

            <a href="dashboard.php">Dashboard</a>
            <a href="#registrations">Latest Registrations</a>
            <a href="#audit">Audit Logs</a>

        <?php } else { ?>

            <a href="student_dashboard.php">Student Dashboard</a>
            <a href="profile.php">My Profile</a>
            <a href="results.php">My Results</a>

        <?php } ?>

        <a href="logout.php">Logout</a>

    </nav>

    <!-- Statistics -->
    <div class="cards">
```

```
        <div class="card">
            <h3>Total Users</h3>
            <h1><?php echo $totalUsers; ?></h1>
        </div>

        <div class="card">
            <h3>Total Students</h3>
            <h1><?php echo $totalStudents; ?></h1>
        </div>

        <div class="card">
            <h3>Total Events</h3>
            <h1><?php echo $totalEvents; ?></h1>
        </div>

    </div>

    <!-- Latest Registrations -->
    <h2 id="registrations">Latest Registrations</h2>

    <table>
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Email</th>
            <th>Role</th>
            <th>Registered Date</th>
        </tr>

        <?php if ($registrations) { ?>

            <?php while ($row = mysqli_fetch_assoc($registrations)) { ?>

                <tr>
                    <td>
                        <?php echo (int)$row['id']; ?>
                    </td>

                    <td>
```

```
                        <?php echo htmlspecialchars($row['name']); ?>
                    </td>

                    <td>
                        <?php echo htmlspecialchars($row['email']); ?>
                    </td>

                    <td>
                        <?php echo htmlspecialchars($row['role']); ?>
                    </td>

                    <td>
                        <?php echo htmlspecialchars($row['created_at']);
?>
                    </td>
                </tr>

            <?php } ?>

        <?php } ?>

    </table>

    <!-- Audit Logs -->
    <h2 id="audit">Recent Audit Logs</h2>

    <table>
        <tr>
            <th>Action</th>
            <th>Date</th>
        </tr>

        <?php if ($auditLogs) { ?>

            <?php while ($log = mysqli_fetch_assoc($auditLogs)) { ?>

                <tr>
                    <td>
                        <?php echo htmlspecialchars($log['action']); ?>
                    </td>
```

```
                    <td>
                        <?php echo htmlspecialchars($log['created_at']);
?>
                    </td>
                </tr>

            <?php } ?>

        <?php } ?>

    </table>

</div>

</body>
</html>
```

Login.php

```
<?php
include 'db.php';

$error = "";

if (isset($_POST['login'])) {
    $email = trim($_POST['email']);
    $password = $_POST['password'];

    $stmt = mysqli_prepare(
        $conn,
        "SELECT id, name, email, password, role
         FROM users WHERE email = ?"
    );

    mysqli_stmt_bind_param($stmt, "s", $email);
    mysqli_stmt_execute($stmt);

    $result = mysqli_stmt_get_result($stmt);
    $user = mysqli_fetch_assoc($result);
```

```
    if ($user &&
        hash_equals($user['password'], hash('sha256', $password))) {

        session_regenerate_id(true);

        $_SESSION['user_id'] = $user['id'];
        $_SESSION['name'] = $user['name'];
        $_SESSION['role'] = $user['role'];
           $stmt = mysqli_prepare(
        $conn,
        "INSERT INTO audit_logs (user_id, action)
         VALUES (?, ?)"
    );

    $action = "Admin logged in";

    mysqli_stmt_bind_param(
        $stmt,
        "is",
        $_SESSION['user_id'],
        $action
    );

    mysqli_stmt_execute($stmt);

        header("Location: dashboard.php");
        exit;

    } else {
        $error = "Invalid email or password";
    }
}
?>

<!DOCTYPE html>
<html>
<head>
    <title>Admin Login</title>
    <link rel="stylesheet" href="style.css">
</head>
```

```
<body>

<div class="box">

    <h2>Admin Login</h2>

    <p class="error">
        <?php echo htmlspecialchars($error); ?>
    </p>

    <form method="POST">

        <input
            type="email"
            name="email"
            placeholder="Enter Email"
            required
        >

        <input
            type="password"
            name="password"
            placeholder="Enter Password"
            required
        >

        <button type="submit" name="login">
            Login
        </button>

    </form>

</div>

</body>
</html>
```

Logout.php

```
<?php
include 'db.php';
```

```
session_unset();
session_destroy();

header("Location: login.php");
exit;
?>
```

Access.php

```
<?php
if (session_status() === PHP_SESSION_NONE) {
    session_start();
}

if (!isset($_SESSION['user_id'])) {
    header("Location: login.php");
    exit;
}

if ($_SESSION['role'] !== 'admin') {
    http_response_code(403);
    exit("Access Denied: Admin only");
}
?>
```

Output



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